

Senzo Ncekana

AI/ML ENGINEER · DATA ANALYST — APPLIED ML, NLP & RISK/COMPLIANCE ANALYTICS

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SUMMARY

AI/ML engineer and data analyst who builds practical AI systems and turns raw, multi-source data into decision-ready insight. Strong in applied machine learning, NLP, and agentic AI (LangChain), alongside Python/SQL analytics, ETL, and anomaly/exception & pattern detection across full data populations. 3+ years in regulated healthcare (compliance, control adherence, and data integrity for audits), now focused on AI engineering and analytics roles in banking and financial services. Comfortable communicating insight to technical and non-technical stakeholders.

TECHNICAL SKILLS

AI / Machine Learning: Applied ML (Supervised & Unsupervised), Neural Networks, NLP, LLMs (Google Gemini, OpenAI), LangChain (Agents), RAG / Vector Search (FAISS).

Data & Analytics: ETL / Data Preparation, Data Warehousing, EDA, Statistical Analysis, Anomaly / Outlier & Exception Detection, Clustering, Classification, Feature Engineering.

Programming & Tools: Python, SQL, SQLite, R, FastAPI, Django, Docker, Git, Relational Databases.

BI & Visualization: Power BI, Tableau, Excel (Pivot Tables, VLOOKUP, Macros), Dashboarding, Data Dictionaries, Reporting, Trend Analysis.

Soft Skills: Stakeholder Engagement · Analytical Thinking · Attention to Detail · Problem-Solving · Managing Multiple Priorities · Comfortable with Ambiguity / Agile.

EXPERIENCE

AI/ML Engineer — Datainno Analytics

Feb 2026 – Present

- Built AI Concierge, an agentic hospitality assistant, using LangChain (tools/agents), OpenAI, and a vector database (Pinecone/PostgreSQL) with RAG to ground responses and reduce hallucinations.
- Engineered LangChain tools that call backend APIs to check availability, process bookings, update room inventory, and log guest interactions — automating end-to-end reservation workflows.
- Integrated the assistant across multiple channels (WhatsApp, Telegram, and web) with intent classification, entity extraction, and context retention for natural, multi-turn conversations.
- Designed an analytics layer surfacing guest-satisfaction sentiment, chat-to-booking conversion, upsell revenue, and peak-demand trends to give hotel management actionable, real-time insight.

AI Engineering Bootcamp — HyperionDev

Feb 2025 – Jan 2026

- Built end-to-end solutions integrating data ingestion, cleaning, transformation (ETL), and predictive modelling.
- Designed reusable, reproducible workflows for structured and unstructured data, emphasising accuracy and data integrity.
- Developed a full-stack Django app with SQL / relational database design; ran EDA and built dashboards to turn data into decision-ready insight.
- Applied supervised/unsupervised ML, neural networks, and NLP; used Git/GitHub for version control and collaboration.

Pharmacy Technician — Nelson Mandela Children's Hospital

Jul 2023 – Present

- Maintain accurate electronic records and inventory data, ensuring data integrity for reporting and audits in a regulated environment.
- Operate within strict regulatory standards, contributing to control adherence and control-effectiveness in a high-volume dispensary.
- Monitor stock and consumption data to detect exceptions and anomalies, supporting formulary and stock-control decisions.
- Analyse workflow data to drive process improvements; collaborate across technical and non-technical stakeholders.

Pharmacy Technician — Local Choice Pelican Pharmacy

Jan 2022 – Jun 2023

- Ensured accurate data entry, record-keeping, and data integrity across dispensary operations while maintaining regulatory compliance.
- Monitored stock trends to flag exceptions and reduced dispensing errors through process improvements.

Pharmacy Technician Intern — Prince Mshiyeni Memorial Hospital

Apr 2021 – Nov 2021

- Monitored inventory levels and prepared reporting to support audits and regulatory compliance.
- Documented patient and stock data accurately; collaborated with healthcare staff to optimise medication processes.

Lab Assistant (Manufacturing) — Nelson Mandela University

Jun 2019 – Nov 2019

- Collected, recorded, and analysed production data to identify efficiency improvements.
- Performed quality-control checks to ensure compliance with safety and quality standards in regulated manufacturing.

PROJECTS

AI Concierge — Agentic Hospitality Assistant (In Progress)

Agentic assistant for hotels handling guest queries and operations (availability, bookings, inventory) via LangChain tools over backend APIs, with RAG to keep responses grounded and reduce hallucinations. **Tech:** LangChain, OpenAI, RAG, Pinecone/PostgreSQL.

PharmaGuard — Pharmaceutical Similarity Detection & Compliance Platform

End-to-end AI platform detecting look-alike/sound-alike (LASA) risks in drug names & packaging; multi-agent pipeline with RAG and OCR/computer vision; full-stack app, Dockerised with CI/CD. **Tech:** Python, FastAPI, Next.js, Google Gemini, LangChain, FAISS, PaddleOCR, PostgreSQL, Docker.

Customer Churn — Risk Scoring ML Pipeline

End-to-end pipeline to score and predict churn (EDA, feature engineering, model comparison, ROC-AUC/F1) — risk/exception scoring across a full population, analogous to flagging high-risk items for control testing. **Tech:** Python, pandas, scikit-learn, XGBoost, Matplotlib, Seaborn.

US Crime Statistics — Pattern & Anomaly Detection

Segmented and detected patterns across a full state-level population using PCA with K-means/hierarchical clustering, with statistical interpretation and visualisation — full-population outlier/exception detection. **Tech:** Python, pandas, scikit-learn, SciPy, Matplotlib, Seaborn.

Amazon Review Sentiment Analysis — Unstructured-Data Analytics at Scale

NLP pipeline over 28,000+ reviews (preprocessing, polarity classification, sentiment scoring, similarity analysis) with reproducible workflows. **Tech:** Python, spaCy, SpacyTextBlob, Jupyter.

Shelf Track — Relational Database & Inventory Management System

Relational database for inventory tracking with full CRUD and search, prioritising data integrity and usability. **Tech:** Python, SQLite, OOP, Database Design.

EDUCATION

AI Engineering Bootcamp — HyperionDev

Feb 2025 – Jan 2026

Machine Learning Fundamentals — Udemy

Nov 2025 – 2026

Diploma in Pharmacy Technical Support — Nelson Mandela University

2017 – 2019

INTERESTS

Passionate about artificial intelligence and emerging technologies, especially large language models, agentic AI, and applied machine learning. Enjoys building open-source and personal side projects to explore new frameworks, sharpen engineering skills, and turn ideas into working systems.